A Mini project on

Eye Ball Cursor Movement Using OpenCV

submitted in partial fulfillment for award of the degree of Bachelor of Technology in Information Technology

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CERTIFICATE

This is to certify that the Mini Project entitled **"Eye Ball Cursor Movement Using Open CV"**

is a bonafide work carried out by

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Abstract

There are different reasons for which people need an artificial of locomotion such as a virtual keyboard. The number of people, who need to move around with the help of some article means, because of an illness. Moreover, implementing a controlling system in it enables them to move without the help of another person is very helpful. The idea of eye controls of great use to not only the future of natural input but more importantly the handicapped and disabled. Camera is capturing the image of eye movement. First detect pupil center position of eye. Then the different variation on pupil position get different command set for virtual keyboard. The signals pass the motor driver to interface with the virtual keyboard itself. The motor driver will control both speed and direction to enable the virtual keyboard to move forward, left, right and stop.

Keywords: Artificial locomotion, Virtual keyboard, Eye movement, Pupil detection.

1. Introduction

1. Introduction

Among the innovations in technology is movement of an eyeball cursor-that is, gaze tracking-with the assistance of which users can control the position of computer cursor movement using eye movements. This field is a promising area in human-computer interaction research, given an impetus from OpenCV-Open-Source Computer Vision Library. One of the most used open-source libraries in real-time computer vision tasks is OpenCV. The facial feature detection application, such as finding out the eyes, is entirely based upon the video stream in real time; therefore, face detection is considered to be the first step with the model of a face- detecting Haar Cascade Classifier towards isolating the eyes belonging to the user. Once the eyes are detected, more complex image processing techniques such as thresholding and contour detection can identify the pupil in the eye. The ability to track movement of the pupil allows the system to determine where the user is looking in space and convey this to a computer cursor movement on the screen. Users can thus maneuver any computer interface solely by the mere movement of gaze. Calibration may be used since it would make accurate cursor control by fine-tuning the mapping with each point that is looked at on the screen. Applications of this technology are very vast. This technology has ranged from assistive technologies for people who have disabilities, making hands-free computer control possible; in gaming and virtual reality applications, this is further enhanced for immersion purposes because users are able to interact with the environment using their eyes. This also provides some critical applications in marketing and usability studies because it helps the researcher understand which area of a screen captures people's most attention. In general, with OpenCV, the movement of eyeball cursors is an exciting new tool in providing for intuitive and hands-free interaction with digital environments. The application of OpenCV, MediaPipe, and PyAutoGUI allows hands-free control of the computer through gaze tracking movements within the eye balls. MediaPipe processes achieve detection and accuracy in facial landmarks and positions of the eyes, through receiving data by OpenCV, which analyzes pupil movement, and finally using PyAutoGUI to translate the gaze information into controlling cursor movements-this makes accessibility as well as interaction a great improvement for different users in their applications. Eye ball cursor movement using OpenCV involves detecting facial features and tracking the pupil's position to translate gaze into cursor movement.

1.1 Purpose of the Project:

OpenCV eye ball cursors will enable the development of hands-free computer interface control through mapping gaze into movement. It has increased access for the disabled because it allows the disabled to move through digital environments without having to rely on physical input devices. The technology of computer vision enables high accuracy levels in tracking eye movements, thereby enhancing the overall user experience. It's also applied in the gaming and virtual reality fields, where intuitive communication through gaze may be correlated with a better and deeper experience. Researchers use this technology in usability studies to analyze the attention of the user and how it is engaged with the interface. Finally, it points to a more universal and engaging approach toward human-computer interaction, thereby opening fields for much needed and innovative applications within industries. This innovation marks a great step toward making technology much more intuitive and user-friendly for everyone.

1.2 Existing System:

Mat lab detects the iris and control curser. Eye movement-controlled wheel chair exists one that controls the wheel chair by monitoring eye movement. In matlab is difficult to predict the Centroid of eye so we go for OpenCV.

1.3 Proposed System:

In our proposed system, computer cursor movement is controlled by eye movements using OpenCV. A camera detects the eye movements, and OpenCV processes the captured images to determine the position of the eye. Based on this, the cursor is controlled to follow the gaze direction. This approach ensures more accurate tracking and allows the use of regular cameras, making the system more practical and efficient.

1.4 Advantages of Proposed System:

- Physically handicapped people can operate computers.
- High Accuracy

2. Related Work

2. Related Work

2.1 Survey:

[1] The public perception of cursor control using Eye Ball Movement OpenCV is complex, reflecting both optimism and concern as awareness of eye-tracking technology increases. On one hand, these systems are viewed as promising tools for improving accessibility, particularly for individuals with disabilities who require hands-free methods of interacting with computers. Eye-tracking systems, including those powered by OpenCV, allow users to control computer cursors through simple eye movements, opening up new possibilities for people with motor impairments who may find traditional input devices like a mouse or keyboard difficult to use. Among this group, there is widespread support for OpenCV-based solutions, which offer a more affordable, customizable alternative to expensive commercial eye-tracking systems. [7] The flexibility of OpenCV, combined with its ability to use standard webcams or affordable cameras, make sit particularly appealing to developers and users seeking low-cost accessibility solutions Beyond assistive technology, another segment of the public sees the potential for eye-tracking to enhance user experiences in gaming and entertainment using MATLAB. Gamers, for instance, are intrigued by the possibility of controlling game environments or interacting with virtual worlds using eye movements, which could create more immersive and intuitive experiences. The integration of eye- tracking technology into virtual reality (VR) and augmented reality (AR) platforms is also generating excitement, as these systems can make navigating complex virtual interfaces more seamless and natural. In this context, OpenCV's open-source nature allows developers to experiment and innovate with different approaches to eye-tracking, potentially leading to new and groundbreaking applications in gaming, entertainment, and beyond. Despite the enthusiasm in certain circles, there are also notable concerns about the accuracy and reliability of eyeball-controlled cursor systems, particularly in everyday use. One of the most frequently cited challenges is the difficulty of achieving precise control with current eye-tracking algorithms. Factors like lighting conditions, head movements, and the natural variation in eye shapes among users can all impact the performance of these systems. Inconsistent tracking can lead to frustration, as the intended gaze direction.

Such real-world conditions. In public forums, users have shared mixed experiences with eye tracking systems, with some praising the potential but others pointing out the practical limitations they encounter during use Another significant issue surrounding [2] eye-tracking Metrices in software, including those systems built with OpenCV, is privacy. Eye-tracking inherently involves the collection of sensitive data related to an individual's gaze patterns, which can reveal detailed information about their interests, intentions, and even emotional states. As the technology becomes more sophisticated and integrated into consumer products, there is growing concern over how this data might be used or misused. People fear that eye tracking could become a tool for surveillance or targeted advertising, with companies or institutions monitoring their gaze behaviors to gather insights without explicit consent. This concern is compounded by the lack of clear regulations governing the use of biometric data, which leaves many users wary of adopting technology that could compromise their privacy. [8] Eye tracking is used to investigate the understandability of cardinal directions. This approach helps assess how effectively individuals comprehend directional cues. Moreover, the utility of eye-tracking for the general public remains a point of debate. While the benefits for people with disabilities are clear, some users question whether eyeball-controlled cursors offer any significant advantage for the average person. Traditional input devices like the mouse, keyboard, and touchscreen are already highly optimized for daily tasks, and for many users, they provide a level of speed and precision that is difficult to match with current eye tracking systems. As such, some members of the public remain unconvinced that this technology will replace or even complement existing devices in everyday scenarios. [6] A real-time eye-tracking system is implemented using an Arduino Uno-based hardware interface. The system captures and processes eye movements in real time. This setup enables accurate and efficient tracking of eye motion. Despite these concerns, public acceptance of eyeball cursor movement systems using OpenCV remains generally positive among specific groups. For individuals with physical disabilities, these systems offer newfound independence and ease of use, enabling them to perform tasks that were previously challenging or impossible. In gaming and entertainment, tech-savvy users and developers are excited by the potential for creating novel interactions and more immersive experiences. The open-source nature of OpenCV also contributes to a sense of optimism, as it allows for experimentation, customization, and community-driven so the

innovation, which may eventually leads to development of more may eventually lead to the development of accurate and userfriendly systems. Looking ahead, the future of eyeball cursor movement systems will likely depend on a few key factors. First and foremost is the need for improvements in accuracy and reliability. Advances in machine learning and computer vision could help create more sophisticated tracking algorithms capable of handling variations in lighting, head position, and user physiology. Additionally, reducing the time and complexity of calibration processes could make these systems more user-friendly, especially for those who are new to the technology. Another important factor will be addressing privacy concerns by establishing clear guidelines and regulations for the collection and use of gaze data, helping to build public trust in the technology. [4] Finally, as the hardware for eye-tracking continues to evolve—such as with higher-resolution cameras and infrared-based tracking systems—the performance of these systems is expected to improve, making them more appealing for a wider range of applications. In conclusion, while public perception of eyeball cursor movement systems using OpenCV is still evolving, there is strong support for their development in specific communities, particularly among people with disabilities and tech enthusiasts. However, broader adoption will depend on overcoming technical challenges related to accuracy, reliability, and privacy.

[3] If these issues can be addressed, eyeball- controlled cursors may become a more integral part of human-computer interaction in the future, offering new ways for people to interact with digital environments hands-free. On one hand, the technology is viewed as a promising tool for improving accessibility, particularly for individuals with physical disabilities, as it allows them to control computer interfaces hands-free. [5] In gaming and entertainment, tech enthusiasts are excited about the potential for immersive experiences, such as using eye movements based human computer interaction to navigate virtual reality environments

. However, there are challenges with accuracy and reliability, as current systems may struggle with factors like lighting conditions and individual eye shapes, leading to frustration for users. Privacy concerns also arise, as eye-tracking collects sensitive gaze data, raising fears of misuse or surveillance While some question the utility of the technology for everyday users, who are already comfortable with traditional input devices like the mouse and keyboard, niche communities see great potential.

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3. Requirement Analysis

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3.1 Functional Requirements:

The functional requirements for the eye ball cursor movement system using OpenCV involve real- time detection and tracking of the user's eyes to translate gaze direction into cursor movement. The system must include a calibration process to map eye movements accurately and allow users to customize sensitivity and speed. It requires input from a standard camera to process video feed for tracking. A clicking mechanism should enable actions through gaze duration or blinking, with adjustable settings for user preferences Load the dataset.

- **Real time video capture.**

Capability of the system to continuously capture and process video frames from a camera at a fast enough rate that it feels immediate to the user, without noticeable lag.

- **Face Detection.**

Face detection is a computer vision technique that identifies and locates human faces in images or videos.

- **Eye Detection.**

Eye detection is a computer vision technique used to locate the eyes within a detected face in image sensor video. It helps in applications like gaze tracking, blink detection.

- **Pupil Detection.**

Pupil detection is the process of identifying the position and size of the pupil in an eye image. This technique is essential for applications such as gaze tracking, assistive technologies, and eye movement analysis.

- **User Interface.**

The user interface (UI) is designed to respond to eye movements by translating gaze direction into cursor actions. The process begins with real-time eye tracking that detects facial landmarks and locates the pupils. Most GUIs include a calibration procedure to accurately map gaze points to cursor positions.

3.2 Non-Functional Requirements:

Non-functional requirements for eye ball cursor and recognition systems focus on the performance, usability, and reliability aspects that ensure the system operates effectively in real- world conditions Response Time

- **Performance:**

The system must offer high processing and low latency, particularly in real time applications like Autonomous driving or live surveillance.
Efficient use of computational resources is crucial to achieve optimal performance with out excessive hardware.

- **Reliability:**

High availability and fault tolerance are essential, ensuring system remains operational under various conditions and recovers gracefully from failures.

- **Accuracy:**

Accuracy is often expressed as the percentage of correct detections or predictions made by the system relative to the total number of attempts. High accuracy indicates that the system can reliably detect eyes and estimate gaze direction.

- **Portability:**

Portability in the context of software systems refers to the ability to run the application across different platforms and devices without requiring extensive modification.

3.3 Computational Requirements:

3.3.1. Software Requirements:

- Operating System : Windows 10 ,11
- Coding Language : python 3.7 or above

3.3.2. Hardware Requirements:

- Processor : Intel i3 and above
- RAM : 4GB and Higher
- Hard Disk : 250 GB

4. Design

4. Design

4.1 Architecture:

4.1.1 System Architecture

The system architecture for eye ball cursor movement using OpenCV is targeted toward real-time gaze tracking with an objective of interactive control of cursors. It starts with an Input Layer, where the camera captures the user's face and streams video, which is actually the primary data source. This video feed is processed in the Preprocessing Module that extracts frames and detects the face of the user using OpenCV's face detection algorithms, including Haar cascade classifiers or deep learning-based techniques.

Once the face is detected, the Feature Detection Module takes control and uses the services of libraries such as MediaPipe to identify important facial landmarks, in this case, the eyes. Here, the system manages to track the exact pupil position through advanced image processing techniques, enabling it to analyze pupils' positions and movements within their eyes accurately.

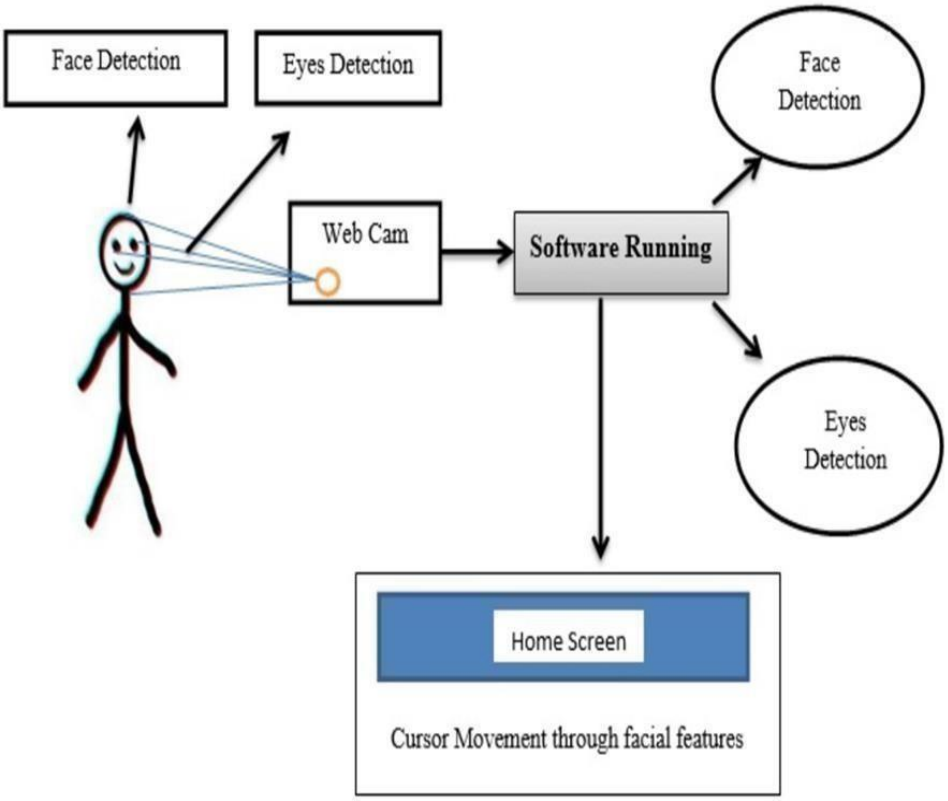


Fig. 4.1.1.1 System Architecture

The Gaze Estimation Module calculates where the user is looking by monitoring pupil movement and includes a calibration routine that maps the gaze points on the screen to exact cursor positions. Calibration is key to ensuring the cursor shows exactly what the user is pointing to, thus improving interaction fidelity.

In the Control Module PyAutoGUI interprets the direction of the gaze and translates it into movement of the cursor on the screen. The users will thus be able to interact with their applications by selecting items or performing actions through eye movements. Fixation detection, which determines when the user is looking at an item long enough to select the same, may be included in the system. Blink detection may be included for clicking.

In order to make the experience richer for the user, a Feedback Mechanism should be used that provides visual feedback, such as highlighted items or messages confirming actions are being executed. Such feedback improves user knowledge about their interactions and increases self-confidence in the system.

The Output Layer finally provides the interface to the users-they may navigate around and control applications using the cursor controlled by a user's gaze. Additionally, the integration of real-time gaze tracking with feedback mechanisms can lead to improved user satisfaction and productivity in tasks requiring prolonged attention.

This technology also holds potential for research in psychology and human-computer interaction, providing insights into attention patterns and user behavior. As the system evolves, advancements in machine learning and computer vision could further refine its accuracy and responsiveness, paving the way for even broader applications.

The future development of augmented reality and virtual reality applications can also include other applications through eye tracking. As these keep gaining speed in technological advancement, eye tracking could make experience more realistic and thus more immersive. The accuracy at which it estimates one's gaze may actually improve over time with interaction since this system utilizes machine learning. It fine-tunes the performance of the system for a given user.

In summary, the eye-ball cursor movement system is a giant leap in human-computer interaction, more than the conventional ways users have been interacting with technology, in a more natural and intuitive way. Next steps in research and development can be expected to unleash innovative applications, improvements in usability, accessibility, and further enhancing interaction design.

4.1.2 Technical Architecture:

The technical architecture of the movement of the eyeball cursor with OpenCV relies on several modules. First, the Input Module captures live video by webcam with OpenCV's Video Capture. Secondly, the Preprocessing Module applies face detection to identify the face and gets the regions of the eyes, using a Haar Cascades or Dlib's 68-point face landmarks model, among others. Using these image processing techniques, including grayscale conversion, Gaussian blurring, and thresholding, on the whole eye region isolated within the Eye Detection and Tracking Module, the entire region of interest undergoes enhancement for better contrast sensitivity. Iris or pupil detection occurs through contour or circular shape detection methods like the Hough Circle Transform that determines the center coordinates of the pupil. This data is input into the Gaze Estimation Module, which renders a map of eye movement relative to the screen space based on geometrical transformations or calibration points. The filters used in the Stabilization Module may include Kalman or low-pass filters to reduce jitter and noise while determining gaze, allowing cursor control to be smooth. Finally, the Control Interface Module is connected to the cursor-control APIs of the operating system in such a manner that they speak to each other. They can beany of PyAutoGUI or c types for Windows. This architecture runs in a continuous loop, processes every video frame in real time, and makes sure that moving the

eyeball does not jerk the cursor.

Fig 4.1.2.1 Technical Architecture

4.2 Algorithm:

The Linear Regression Algorithm can be applied to map the relationship between the eye's pupil position and the corresponding cursor position on the screen. This involves establishing a linear relationship between the coordinates of the detected pupil (input) and the screen coordinates (output), enabling smooth and accurate cursor movement. The system first captures the pupil coordinates (let's denote them as) in the camera frame, typically using image processing techniques like thresholding and contour detection. The corresponding screen coordinates of the cursor (denoted as are determined based on the user's gaze. Using linear regression, the relationship between these two sets of coordinates can be modeled as:

$$x_{\text{cursor}} = \beta_0 + \beta_1 x_{\text{pupil}} + \epsilon$$

$$y_{\text{cursor}} = \alpha_0 + \alpha_1 y_{\text{pupil}} + \epsilon$$

Where

:

- β0 and α0 are the intercepts (cursor position when the pupil is at the origin),
- β1 and α1 are the slopes (mapping the scale of eye movement to screen movement),
- ϵ represents the error terms.

However, linear regression could conceptually enhance such systems by modeling the relationship between gaze positions on the screen and eye landmark coordinates detected in real time. By collecting data pairs of eye positions and corresponding screen coordinates, a linear regression model could be trained to predict cursor locations based on new eye movement data.

This training process would involve finding the best-fit line that minimizes the error between predicted and actual screen positions, thereby improving cursor accuracy. Additionally, a dynamic approach could allow the model to update its parameters continuously, adapting to variations in user behavior or differences in eye physiology. While the current implementation does not incorporate linear regression, integrating this technique could significantly refine the mapping of eye movements to cursor positions, resulting in a more precise and responsive user experience.

5. Implementation

5.Implementation

5.1 Technologies:

In order to achieve eye-ball mouse movement control using OpenCV, we will require an integrated development environment, PyCharm, that will ease the actual coding as well as testing process for an application developer. By having this type of IDE, developers will simply be able to maintain control over their codes, debug the issues, and run their tests in a relatively friendly environment. In the development of the project, there are various major libraries involved based on the purpose they serve and their added contribution to the enrichment of the functionality of the system.

OpenCV happens to be the core component of the project and provides robust solutions towards image processing and computer vision operation. It can be used to detect facial features and the position of the eyes in real time, and thus, the accurate eye tracking is feasible. MediaPipe supports this process by using efficient models of facial landmarks detections that enhance the precision associated with eye tracking. This is used for the system to locate and point toward the pupil with incredible accuracy.

We make use of the PyAutoGUI library to convert raw gaze data into cursor movements. Once such a set of information is interpreted by PyAutoGUI, it then becomes feasible for people to control the computer through movements of their cursor; it opens up new possibilities for all people with impairments to use their computer interface partly or completely hands-free.

This advanced eye-tracking technology has applications beyond accessibility. For instance, in gaming and virtual reality, it brings immersive experiences, where the user interacts directly with the environment by looking at objects. As a whole, combining PyCharm, OpenCV, MediaPipe, and PyAutoGUI, this system is intuitive and responsive. The new approach really changes how users navigate and control digital environments. Therefore, this innovative approach is one giant leap for human computer interaction, which will make technology much more accessible and user-friendly for everyone.

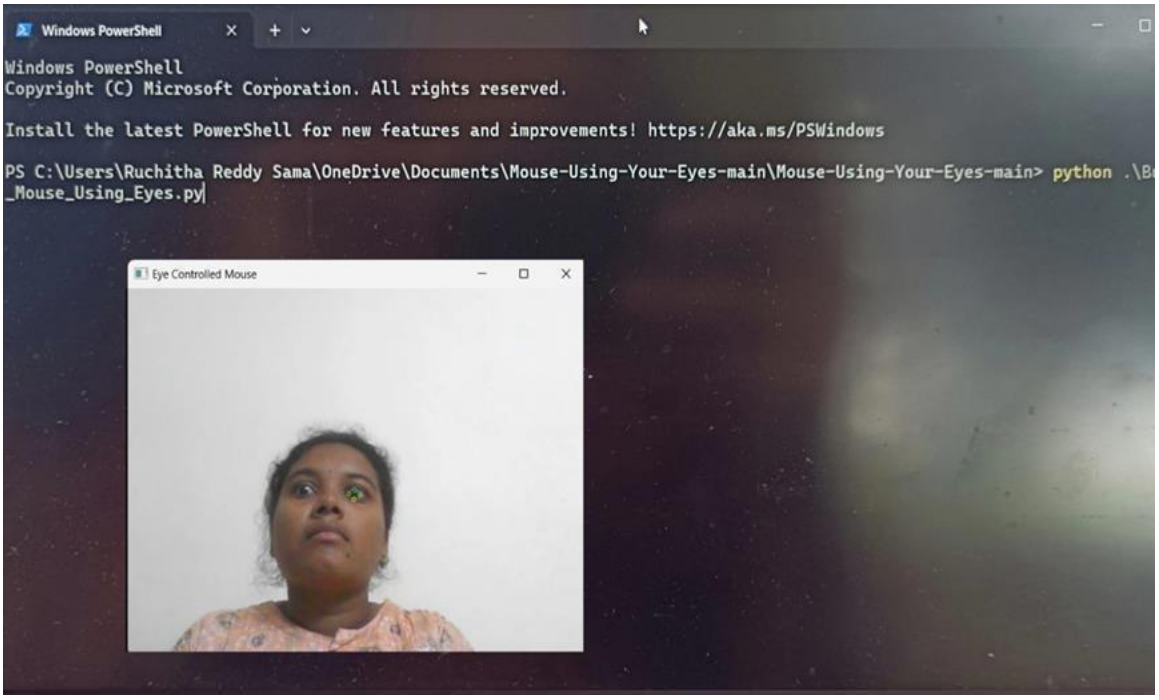
5.2 Pseudo code

```
import cv2
import mediapipe as mp
import pyautogui
    cam = cv2.VideoCapture(0)
face_mesh = mp.solutions.face_mesh.FaceMesh(refine_landmarks=True)
screen_w, screen_h = pyautogui.size()
    while True:
  _, frame = cam.read()
frame = cv2.flip(frame, 1)
rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
output = face_mesh.process(rgb_frame)
  landmark_points = output.multi_face_landmarks frame_h,
frame_w, _ = frame.shape
if landmark_points:
  landmarks = landmark_points[0].landmark
  for id, landmark in enumerate(landmarks[474:478]):
    x = int(landmark.x * frame_w)
  y = int(landmark.y * frame_h)
  cv2.circle(frame, (x, y), 3, (0, 255, 0)) if
  id==1:
  screen_x = screen_w / frame_w * x
   screen_y = screen_h / frame_h * y
   pyautogui.moveTo(screen_x, screen_y)
    left = [landmarks[145], landmarks[159]]
   for landmarks in left:
          x = int(landmark.x * frame_w)
           y = int(landmark.y * frame_h)
          cv2.circle(frame, (x, y), 3, (0, 255, 255))
 if (left[0].y - left[1].y) < 0.004:
```

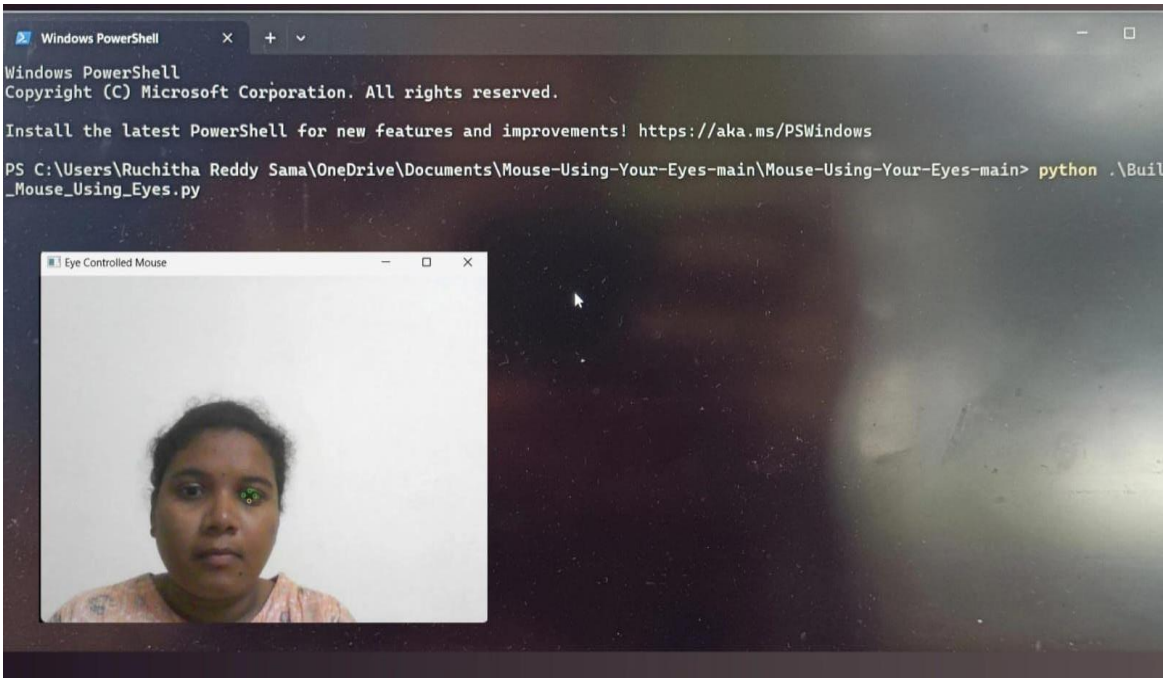
```
    pyautogui.click()
     pyautogui.sleep(1)
cv2.imshow('Eye Controlled Mouse', frame)
cv2.waitKey(1).
```

6. Screenshots

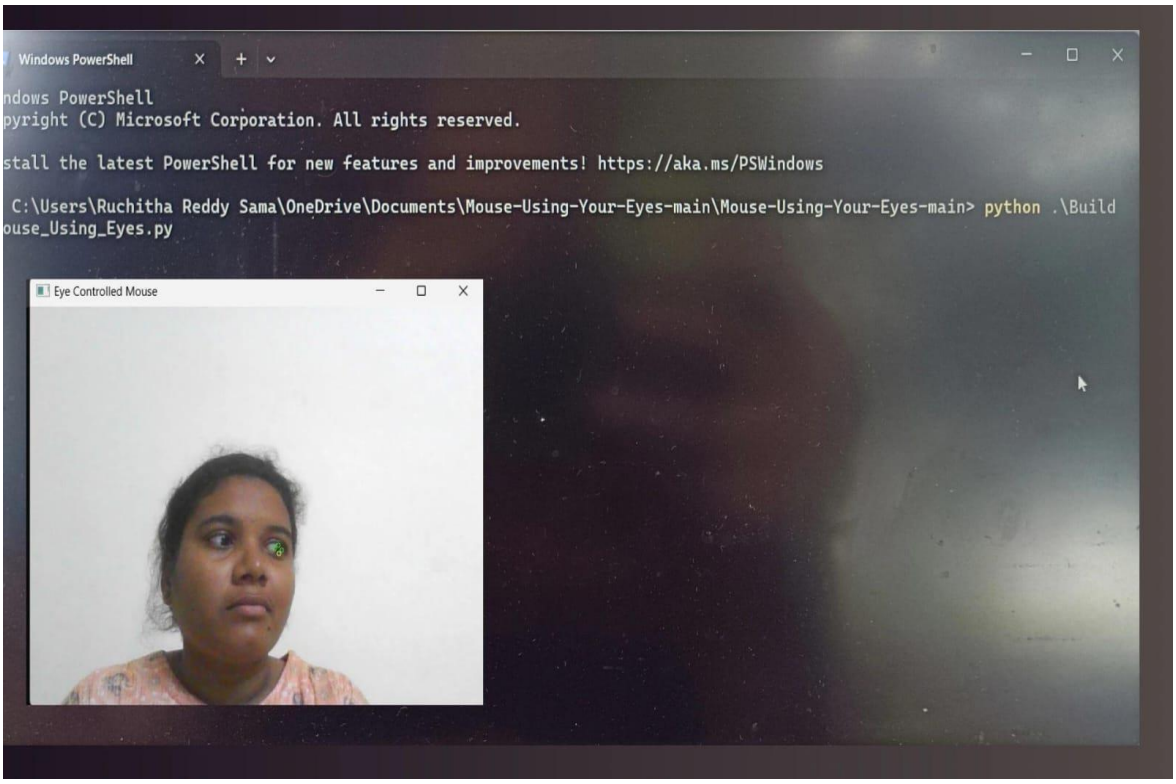
6. Screenshots



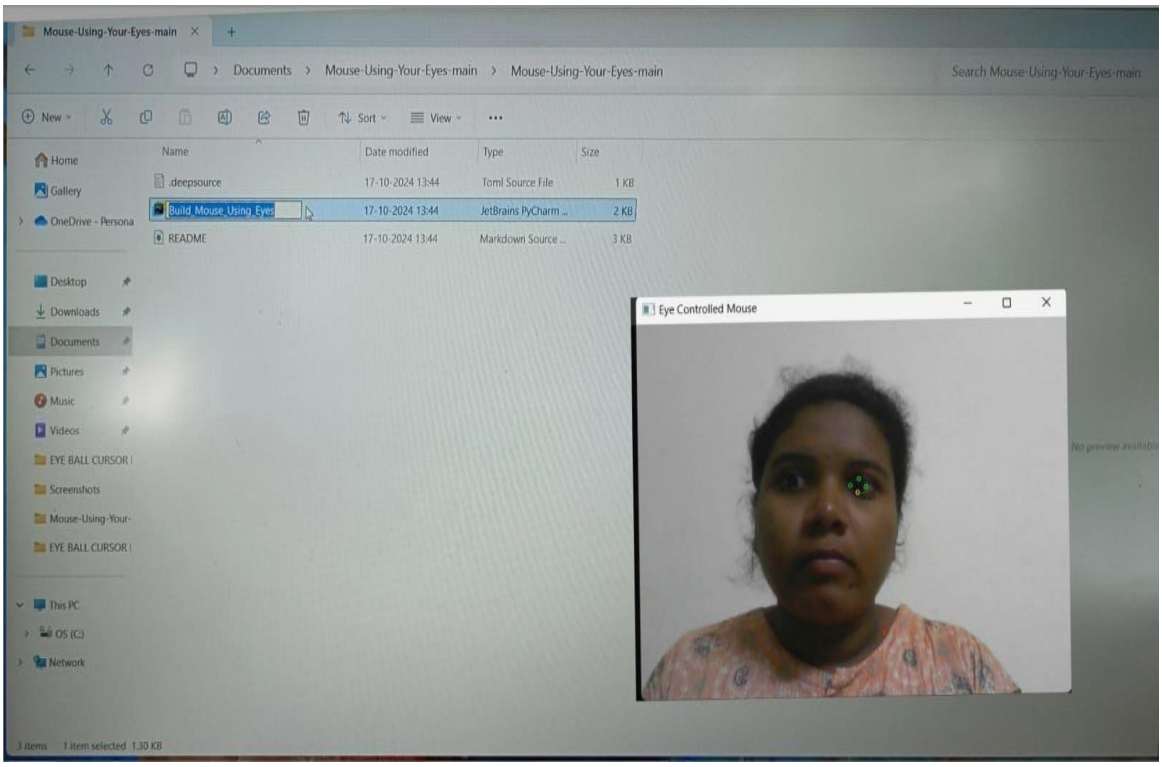
Screenshot 6.1 cursor moving upwards



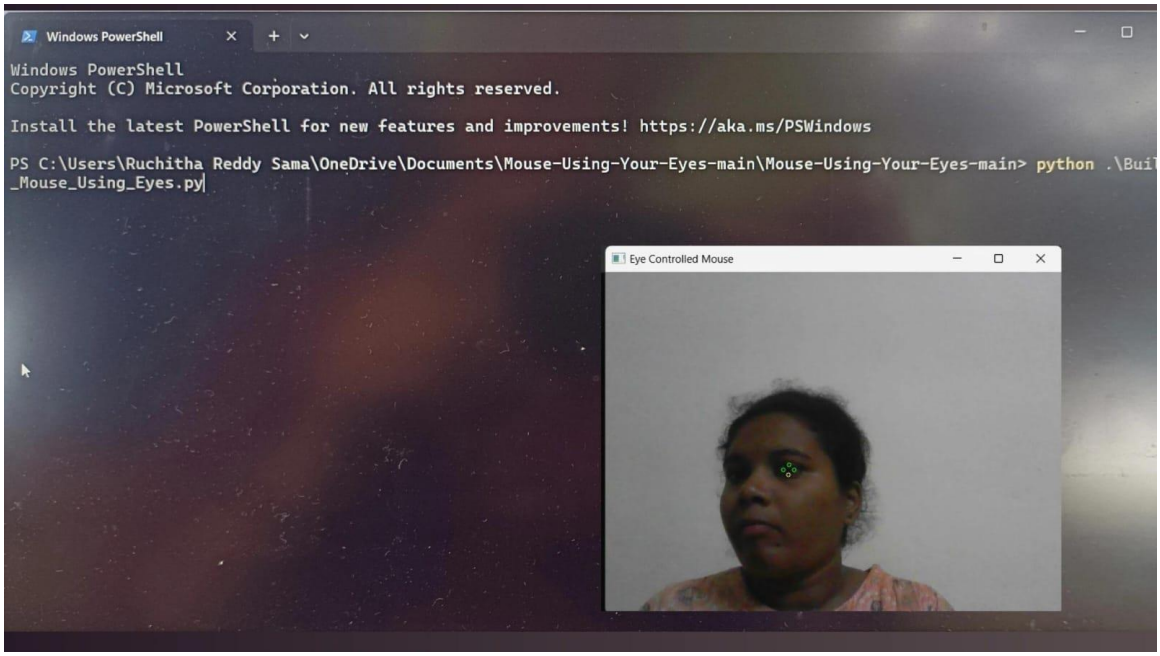
Screenshot 6.2 cursor at the canter



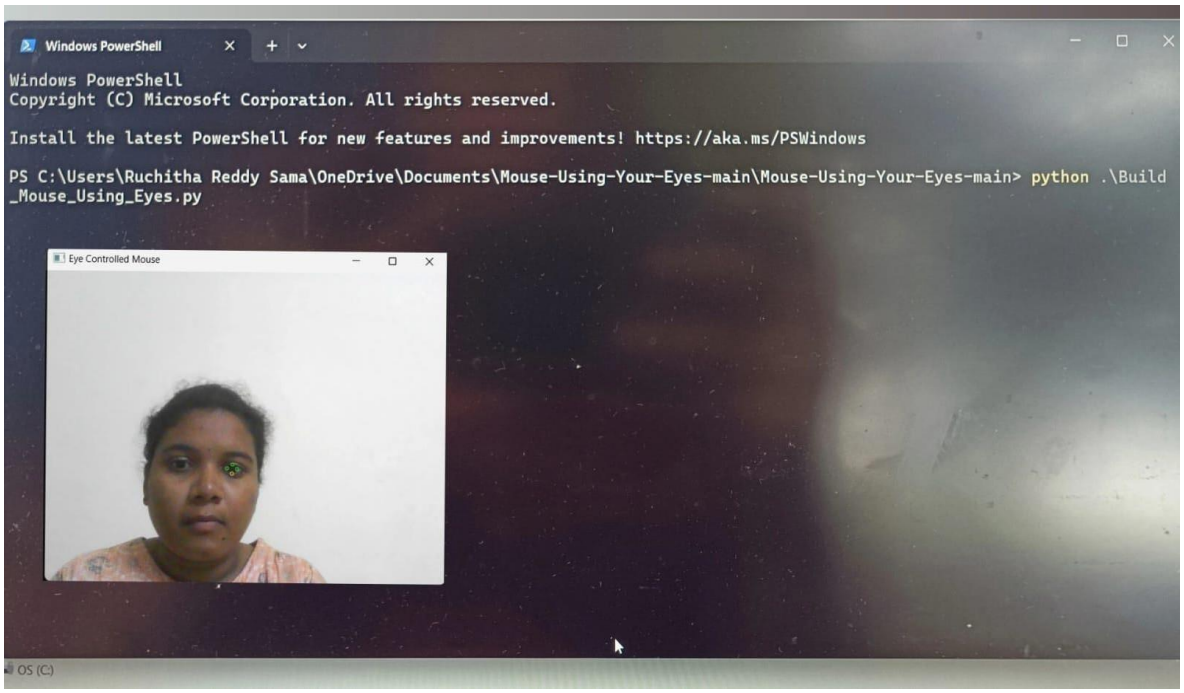
Screenshot 6.3 Cursor moving towards right side



Screenshot 6.4 Eye Blinking as a Click Mechanism



Screenshot 6.5: Cursor moving towards left side



Screenshot 6.6: Cursor moving downwards

7. Test Cases

7. Test Cases

S.no	Test Case	Input	Expected Output	Actual Output	Result
1	Detect face and eyes	Webcam stream of a person looking straight at the camera	Eyes detected within face bounding box	Eyes detected correctly	pass
2	Track left eye movement	Webcam stream, user looking left	Cursor moves left	Cursor moves left	Pass
3	Track right eye movement	Webcam stream, user looking right	Cursor moves right	Cursor moves right	Pass
4	Track upward eye movement	Webcam stream, user looking upward	Cursor moves upward	Cursor moves upward	Pass
5	Track downward eye movement	Webcam stream, user looking downward	Cursor moves downward	Cursor moves downward	Pass
6	Blink detection	Webcam stream, user blinks	Cursor stops/pauses	Cursor stops correctly	Pass
7	No face in the frame	Webcam stream with no person	No eye detection, no cursor movement	No eye detection, no cursor movement	Pass
8	Partial face obstruction	Webcam stream with user's face partially obstructed	Partial eye detection, limited or no cursor movement	Partial eye detection, limited or no cursor movement	Pass
9	Lighting condition variation	Webcam stream with dim/bright lighting	Consistent eye detection and cursor movement	Detection was affected, inconsistent cursor movement	Fail
10	Multiple faces in the frame	Webcam stream with two people in frame	Only one face should be tracked	Both faces detected, inconsistent cursor behaviour	Fail

8. Conclusion

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Eye ball cursor movement using OpenCV represents a significant advancement in human-computer interaction, enabling users to control their devices hands-free through intuitive eye movements. By leveraging powerful tools like OpenCV and Media Pipe, developers can implement effective systems for real-time eye and gaze tracking, enhancing accessibility and user experience across various applications. The process involves critical steps, including face detection, eye detection, and gaze estimation, all of which work together to translate the user's visual focus into accurate cursor movement on the screen. This technology not only opens doors for individuals with mobility impairments but also enhances interaction in gaming, virtual reality, and other immersive environments. As the field of computer vision continues to evolve, the integration of eye tracking with advanced algorithms and machine learning will further refine cursor control, making it more responsive and intuitive. Ultimately, eye ball cursor movement stands as a testament to the potential of combining innovative technology with practical applications, paving the way for a more inclusive and interactive digital future.

9. Future Scope

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The future scope of eye ball cursor movement using OpenCV is promising, with significant potential for innovation and improvement. Enhancing accuracy through better calibration techniques and adaptive machine learning algorithms can optimize cursor control based on individual users' eye characteristics. Additionally, integrating eye tracking with augmented and virtual reality can create more immersive experiences, allowing for natural interactions in virtual spaces. Expanding applications in accessibility will further assist individuals with disabilities in navigating technology, while combining eye tracking with voice commands or gesture recognition can offer a multi-modal interaction approach. Real-time analytics can provide valuable feedback on user behavior, particularly in gaming, to enhance engagement. Lastly, integrating eye tracking into mobile and wearable devices will enable.

10. References

10. References

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